UNIT 2 · FORCES & NEWTON'S LAWS · WEEK 2

Problem Set 8 — Third Law & Friction

Physics 2026–27 · due at the start of class, Monday, November 16

NAME _____ DATE _____

How to work this set. Friction template, every time: find N from vertical equilibrium (it equals mg *only* when nothing else pushes vertically), then $f_s \le \mu_s N$ or $f_k = \mu_k N$, then $\Sigma F = ma$. Remember the ≤: static friction is a *ceiling*, not a fixed value.

1 Name the reaction

THIRD LAW

For each action force, name the reaction: magnitude, direction, and — most importantly — *which object it acts on*.

a. A swimmer's hand pushes water backward.

b. Earth's gravity pulls down on a falling mango.

c. A hammer strikes a nail, pushing it downward.

2 Why don't they cancel?

THIRD LAW

A classmate argues: *"If every force has an equal and opposite reaction, the forces always cancel, so nothing should ever accelerate."* In two or three sentences, find the flaw. (The key word is **objects**.)

ANSWER

3 Two coefficients, two numbers

FRICTION

A crate weighing 300 N sits on a level floor with $\mu_s = 0.40$ and $\mu_k = 0.30$. (Note: you're given the *weight*, not the mass.)

a. Find the maximum static friction force.

b. Find the kinetic friction force once it's sliding.

c. Which is bigger, and what does that mean physically about starting vs. keeping a slide going?

WORK

4 Does it move?

STATIC FRICTION

A 15 kg toolbox sits on a floor with $\mu_s = 0.45$. You push horizontally with 60 N.

a. Run KDEA: find $f_{s,max}$ and compare it to your push. Does the toolbox move?

b. Trick part: what is the actual friction force acting on the toolbox right now? (Careful — it is *not* $f_{s,max}$.)

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	N = MG HERE — SAY WHY THAT'S ALLOWED · $F_s,MAX = M_sN$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Sliding with friction

KINETIC FRICTION

A 30 kg equipment case is already sliding across the gym floor ($\mu_k = 0.25$) while being pushed horizontally with 120 N. Draw the FBD, then run KDEA for the acceleration.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	N, THEN $F_k = M_kN$, THEN $\Sigma F = F_a - F_k = MA$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

6 The angled pull

CHALLENGE

A `10 kg` box is dragged across a floor (μ_k `= 0.20`) by a rope pulling with `40 N` at `30°` above horizontal.

a. Find N. (The rope's upward component is holding part of the weight — N is *less* than mg.)

b. Find the friction force.

c. Find the acceleration.

d. One sentence: why did pulling at an angle change the friction, when nothing about the floor changed?

K	EVERY GIVEN, WITH UNITS · DECOMPOSE T FIRST
D	THE TARGETS & THEIR UNITS
E	VERTICAL: N = MG − T SIN0 · HORIZONTAL: ΣF = T COS0 − M_kN = MA
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

Graded on the homework 4–3–2–1 scale. Problem 4b is the week's trap: static friction supplies exactly what's needed to balance the push — the ceiling number is only the breaking point.